

MANAS RANJAN DAS

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SUMMARY

Electrical and Computer Engineering student & Researcher with 3+ years of experience in **Full-Stack Development, AI/ML, and IoT systems**. Specialized in building intelligent hardware-software interfaces and scalable cloud architectures. Proven track record in optimizing high-complexity codebases and conducting deep-tech research in computer vision, energy harvesting, and microgrid stability.

EDUCATION

Biju Patnaik University of Technology, Rourkela
Bachelor of Technology in Electrical & Computer Engineering

2023 — 2027
CGPA: 8.82 / 10.00

Jatiya Kabi Bira Kishore Das Govt. HSS
Higher Secondary Science (CHSE Odisha)

2021 — 2023
87.5%

WORK EXPERIENCE

AI Engineering Intern | Odisha Computer Application Centre (OCAC) **May 2025 — July 2025**

- Developed **YOLO Deep Learning based PCB Component Detector** and OpenCV gesture recognition systems (Writing Pad, On-Screen Keyboard).
- Engineered a **Retail-Sales-Forecast Model** using ML with a frontend dashboard for real-time inventory prediction.
- Built browser extensions for Email-Spam filtering and text-based Emotion Detection using ML pipelines.

Research Intern | HBCSE, Tata Institute of Fundamental Research (TIFR) **June 2024 — Present**

- Developing educational research-informed resources for high school physics in a **two-year research tenure**.
- Designing **interactive simulations** to visualize abstract physics concepts for nationwide distribution.

Software Development & Solution Architecture Intern | Accenture / AWS **Dec 2024**

- Optimized Sims 4 legacy codebase** using advanced data structures, reducing memory overhead by **20%**.
- Designed scalable hosting architecture on **AWS Elastic Beanstalk**, reducing response latency by 30% for high-growth clients.

SELECTED PROJECTS

AI-Powered PCB Component Detector | Python, OpenCV, YOLOv8 **2025**

- Engineered a deep learning-based recognition system for manufacturing QA; achieved high precision in real-time labeling and detection of small-form-factor PCB components using custom **YOLO** architectures.

C-- Scripture Language | Python, React, Flask, C **2026**

- Developed a browser-based **Live Script Notebook** for C programming, featuring a custom compilation pipeline that enables real-time execution and visualization of C code within a React frontend that basically provides output through V8 engine.

Project Income - "ShineX" | MERN Stack, Vercel, Render **2025**

- Architected a full-stack E-Commerce platform for a utility brand using the **MERN stack**; integrated secure authentication, real-time inventory management, and a responsive UI/UX.

Hybrid IoT Energy Harvesting System | Arduino, IoT, R&D **2024**

- Designed an integrated node harvesting energy from **Solar (60%), Piezo (30%), and RF (10%)**; presented research at the IoT National Conclave.

TECHNICAL SKILLS

Languages: Python, C, C++, JavaScript (ES6+), SQL, R, PHP, Scratch
Web/Dev: React, Node.js, Express, Flask, Django, MongoDB, Docker, Git/GitHub, Bootstrap
AI/Data: TensorFlow, Scikit-Learn, YOLO, OpenCV, NLP, MATLAB, Advanced Excel
Hardware/IoT: Arduino, Raspberry Pi, Embedded Systems, Power Electronics, TinkerCad, Robotics

ACHIEVEMENTS & LEADERSHIP

- Membership:** Member, Indian Society of Technical Education (ISTE) from 2024 - 2028.
- BPUT Tech Carnival 2025:** Secured **State Level Silver Medal** in Math Olympiad and **Bronze Medal** in Technical Quiz among 150+ top tech colleges.
- Research Publication:** Co-authored "Optimizing fuzzy power system stabilizers..." in **Engineering Research Express**.
- NIUS 2024 Scholar:** Selected as 1 of 71 nationwide for **NIUS Physics Program at HBCSE-TIFR**.
- Social Entrepreneurship:** CEO of Aerial.Vue (Student Start-up) and Co-Founder of InceptriX (AI/ML Solutions).
- Competition:** Secured **AIR 154,047** in JEE Mains 2023 among 1 Million+ candidates.
- Certifications:** Adobe Certified Content Specialist, MBA (Germany), JUNO Mastered ChatGPT.
- Leadership:** Class Representative (ECE 23-27); Successfully Led 5 Teams for student-led innovation challenges, BPUT-Hackathon 2025.